Determinants, $\vec{\S}$2.3

Recall from the last few classes:

- We define the **determinant** of $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, denoted by $\det(A)$, by

$$\det(A) = ad - bc.$$

- **Theorem:** If A is an n by n invertible matrix, then for each $\vec{b} \in \mathbb{R}^n$, $A\vec{x} = \vec{b}$ has a unique solution; namely $\vec{x} = A^{-1}\vec{b}$.

- **Theorem:** Suppose A and B are n by n invertible matrices, then the following hold.

  1. A^{-1} is invertible and $(A^{-1})^{-1} = A$.
  2. $(AB)^{-1} = B^{-1}A^{-1}$.
  3. $(A^T)^{-1} = (A^{-1})^T$.

- An **elementary matrix** E is a matrix obtained from the identity matrix by a single elementary row operation.

- **Theorem:** A is invertible if and only if A is row equivalent to the identity matrix. Furthermore, any sequence of elementary row operations that reduces A to I_n also transforms I_n into A^{-1}.

1.

(a) Let $A = \begin{bmatrix} 1 & -2 & -1 \\ -1 & 5 & 6 \\ 5 & -4 & 5 \end{bmatrix}$. Find A^{-1} if possible.

(b) Let $A = \begin{bmatrix} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{bmatrix}$. Find A^{-1} if possible.

Invertible Matrix Theorem Let A be an $n \times n$ matrix, then the following are all equivalent.

(a) A is invertible.

(b) A is row-equivalent to I_n.

(c) A has n pivot positions.

(d) $A\vec{x} = \vec{0}$ has only the trivial solution.

(e) The columns of A are linearly independent.

(f) $x \mapsto A\vec{x}$ is one-to-one (injective).

(g) For all $\vec{b} \in \mathbb{R}^n$, $A\vec{x} = \vec{b}$ is consistent.

(h) The columns of A span $\mathbb{R}^n$.

(i) $\vec{x} \mapsto A\vec{x}$ is onto (surjective).

(j) There exists a matrix C such that $CA = I_n$.

(k) There exists a matrix D such that $AD = I_n$.

(l) A^T is invertible.

2. Prove the theorem.

3. Determine if the following matrices are invertible.

(a) Let $A = \begin{bmatrix} 3 & 0 & 0 \\ -3 & -4 & 0 \\ 8 & 5 & -3 \end{bmatrix}$.

(b) Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

Theorem: Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be linear and let A be the associated matrix. Then T is invertible if and only if A is invertible.

4. Prove the theorem.

Goal: Develop a numerical test for invertibility.

5. Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$. Find a condition on the entries of A for A to be invertible.

Definition 1. *For a matrix A define A_{ij} to be the matrix obtained from A by deleting row i and column j.*

Remark:

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = a\text{det}A_{11} - b\text{det}A_{12} + c\text{det}A_{13}.$$

Definition 2. *For $n \geq 2$, the determinant of an $n \times n$ matrix $A = [a_{ij}]$ is*

$$\det(A) = a_{11}\det(A_{11}) - a_{12}\det(A_{12}) + \cdots + (-1)^{n+1}a_{1n}\det(A_{1n}) = \sum_{j=1}^{n}(-1)^{1+j}a_{1j}\det(A_{1j}).$$

6. Compute the determinant of $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

Definition 3. *The (i, j)-**cofactor** of A is given by $c_{ij} = (-1)^{i+j}\det A_{ij}$, and the **cofactor expansion along row** i is given by $\sum_{j=1}^{n} a_{ij}c_{ij}$.*

Theorem: Cofactor expansion across any row or column is the same, and hence each cofactor expansion gives the determinant.

7. Compute the determinant of $C = \begin{bmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{bmatrix}$.

Definition 4. *Let $A = [a_{ij}]$ be an $n \times n$ matrix, then A is* **upper triangular** *if $a_{ij} = 0$ when $i > j$.*

Theorem: Let A be an $n \times n$ upper triangular matrix, then

$$\det(A) = a_{11}a_{22}\dots a_{nn} = \prod_{j=1}^{n} a_{jj}.$$

8. Prove the theorem.

Determinants – Answers / Solutions

1. (a) Let $A = \begin{bmatrix} 1 & -2 & -1 \\ -1 & 5 & 6 \\ 5 & -4 & 5 \end{bmatrix}$. We try augment A by I_n and try to use the result that if A is invertible, then $[A|I_n] \sim [I_n|A^{-1}]$. Thus

$$\left(\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ -1 & 5 & 6 & 0 & 1 & 0 \\ 5 & -4 & 5 & 0 & 0 & 1 \end{array}\right) \sim \left(\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ 0 & 3 & 5 & 1 & 1 & 0 \\ 0 & 6 & 10 & -5 & 0 & 1 \end{array}\right) \sim \left(\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ 0 & 3 & 5 & 1 & 1 & 0 \\ 0 & 0 & 0 & -7 & -2 & 1 \end{array}\right),$$

and now we're stuck. Since A is not row-equivalent to I_n (which we're seeing on left), the algorithm fails to find an inverse for A. Thus we know A is not invertible.

(b) Let $A = \begin{bmatrix} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{bmatrix}$. We again augment A by I_n and try to use the result that if A is invertible, then $[A|I_n] \sim [I_n|A^{-1}]$. Thus

$$\begin{aligned} \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ -3 & 1 & 4 & 0 & 1 & 0 \\ 2 & -3 & 4 & 0 & 0 & 1 \end{array}\right) &\sim \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & -3 & 8 & -2 & 0 & 1 \end{array}\right) \\ &\sim \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & 0 & 2 & 7 & 3 & 1 \end{array}\right) \\ &\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 8 & 3 & 1 \\ 0 & 1 & 0 & 10 & 4 & 1 \\ 0 & 0 & 2 & \frac{7}{2} & \frac{3}{2} & \frac{1}{2} \end{array}\right), \end{aligned}$$

and hence $A^{-1} = \begin{bmatrix} 8 & 3 & 1 \\ 10 & 4 & 1 \\ \frac{7}{2} & \frac{3}{2} & \frac{1}{2} \end{bmatrix}$.

2. *Proof.* There are many ways to organize all the implications. Here I try to use previous results from class as much as possible.

$(a) \implies (j)$ by definition.

$(j) \implies (d)$ by Theorem 2.5 (or above on this handout).

$(d) \implies (c)$ by Theorem 1.4.

$(c) \implies (b)$ since A is square.

$(b) \implies (a)$ by Theorem 2.7 (or above on this handout).

Thus we have established $(a) \iff (b) \iff (c) \iff (d) \iff (j)$.

Note $(a) \implies (k)$ by definition of invertibility. We prove $(k) \implies (g)$. Suppose D is an $n \times n$ matrix such that $AD = I_n$. Let $\vec{b} \in \mathbb{R}^n$, and define $\vec{y} = D\vec{b}$. Now $A\vec{y} = AD\vec{b} = I_n\vec{b} = \vec{b}$, and hence $\vec{y}$ is a solution to the matrix equation $A\vec{x} = \vec{b}$. Thus the matrix equation is always consistent, and hence (g) follows. Now we prove $(g) \implies (a)$. By Theorem 1.4, $A\vec{x} = \vec{b}$ being consistent for any $\vec{b}$ is equivalent to A having a pivot position in each row. Since A is square, we know A must be row equivalent to the identity matrix. By the theorem above, A is invertible, and hence (a) holds.

Finally note $(d) \iff (e) \iff (f)$ and $(g) \iff (h) \iff (i)$ follows from Theorem 1.12. Finally we prove $(a) \iff (l)$ above on this handout. $\square$

3. (a) We could row-reduce A to determine if A is row equivalent to the identity matrix, but in this case we can see the columns of A are linearly independent, and hence we know A must be invertible from the invertible matrix theorem.

(b) We have previously seen this matrix does not span all of $\mathbb{R}^3$, and hence we can use the invertible matrix theorem to conclude A is not invertible.

4. *Proof.* We prove both implications directly from the definitions.

$\implies$ Suppose T is invertible, then there exists a function $T^{-1} : \mathbb{R}^n \to \mathbb{R}^n$ such that $T(T^{-1}(\vec{x})) = \vec{x}$ and $T^{-1}(T(\vec{x})) = \vec{x}$ for all $\vec{x} \in \mathbb{R}^n$. If we knew that T^{-1} was linear, then it would have an associated matrix we could use.

Claim: T^{-1} is linear.

Proof of Claim: Let $\vec{x}, \vec{y} \in \mathbb{R}^n$ and $c \in \mathbb{R}$. Since T is invertible, it is bijective. Thus there exists unique $\vec{a}, \vec{b} \in \mathbb{R}^n$ such that $T(\vec{a}) = \vec{x}$ and $T(\vec{b}) = \vec{y}$. Now we check the conditions for linearity:

$$T^{-1}(\vec{x}+\vec{y}) = T^{-1}(T(\vec{a}) + T(\vec{b})) = T^{-1}(T(\vec{a}+\vec{b}) = \vec{a} + \vec{b} = T^{-1}(\vec{x}) + T^{-1}(\vec{y}).$$

and

$$T^{-1}(c\vec{x}) = T^{-1}(cT(\vec{a})) = T^{-1}(T(c\vec{a})) = c\vec{a} = cT^{-1}(\vec{x}).$$

Thus T^{-1} is linear.

Since T^{-1} is linear, it has an associated matrix D. Now

$$AD\vec{x} = T(T^{-1}(\vec{x})) = \vec{x}$$

and

$$DA\vec{x} = T^{-1}(T(\vec{x})) = \vec{x}.$$

Thus D is the inverse matrix of A, and hence A is invertible.

$\impliedby$ Suppose A is invertible. Define $S : \mathbb{R}^n \to \mathbb{R}^n$ by $S(\vec{x}) = A^{-1}\vec{x}$. Then $T(S(\vec{x})) = AA^{-1}\vec{x} = \vec{x}$ and $S(T(\vec{x})) = A^{-1}A\vec{x} = \vec{x}$. Thus S is the inverse function of T, and hence T is invertible. $\square$

5. By the invertible matrix theorem, we know A is invertible if and only if A is row equivalent to the identity matrix. Thus we try to row-reduce A and see what condition we get on the entries of the matrix.

If the first column of A is all zeros, then we know the matrix is not invertible. Thus we assume $a \neq 0$.

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \sim \begin{bmatrix} a & b & c \\ ad & ae & af \\ ag & ah & ai \end{bmatrix} \sim \begin{bmatrix} a & b & c \\ 0 & ae-bd & af-cd \\ 0 & ah-bg & ai-cg \end{bmatrix}$$
$$\sim \begin{bmatrix} a & b & c \\ 0 & ae-bd & af-cd \\ 0 & (ah-bg)(ae-bd) & (ai-cg)(ae-bd) \end{bmatrix}$$
$$\sim \begin{bmatrix} a & b & c \\ 0 & ae-bd & af-cd \\ 0 & 0 & D \end{bmatrix}$$

where $D = (ai - cg)(ae - bd) - (af - cd)(ah - bg)$.

Now multiplying out D and simplifying we get

$$D = a(aei - ceg - bdi - afh + cdh + bfg).$$

Since a is non-zero, we know for A to be row equivalent to the identity matrix, we need $(aei - acg - bdi - afh + cdh + bfg) \neq 0$. We define

$$\det(A) = (aei - acg - bdi - afh + cdh + bfg).$$

This seems like a strange condition, so let's play around with it a bit more. Factoring we get the following

$$(aei - acg - bdi - afh + cdh + bfg) = a(ei - fh) - b(di - fg) + c(dh - eg),$$

and now we see some smaller determinants showing up; namely

$$a(ei-fh)-b(di-fg)+c(dh-eg) = a\cdot\det\begin{bmatrix} e & f \\ h & i \end{bmatrix} - b\cdot\det\begin{bmatrix} d & f \\ g & i \end{bmatrix} + c\cdot\det\begin{bmatrix} d & e \\ g & h \end{bmatrix} = \det(A).$$

This expression for the determinants suggests a way to define the determinant of an $n \times n$ matrix in terms of $(n-1)\times(n-1)$ matrices.

6. We use the definition to compute the determinant:

$$\det A = \det\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = 1\cdot\det\begin{bmatrix} 5 & 6 \\ 8 & 9 \end{bmatrix} - 2\cdot\det\begin{bmatrix} 4 & 6 \\ 7 & 9 \end{bmatrix} + 3\cdot\det\begin{bmatrix} 4 & 5 \\ 7 & 8 \end{bmatrix}$$
$$= 1(45-48) - 2(36-42) + 3(32-35) = 0.$$

We have previously seen that A is not invertible, so it is reassuring to see the determinant is also zero.

7. Again we use the definition to compute the determinant:

$$\det C = \det \begin{bmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{bmatrix} = 6 \cdot \det \begin{bmatrix} 7 & 2 & -5 \\ 0 & 0 & 0 \\ 3 & 1 & 8 \end{bmatrix} - 5 \cdot \det \begin{bmatrix} 1 & 7 & 2 \\ 2 & 0 & 0 \\ 8 & 3 & 1 \end{bmatrix} = -5(-2) = 10.$$

Using the last Theorem, we can expand about the third row as well. We show that here:

$$\det C = \det \begin{bmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{bmatrix} = 2 \cdot \det \begin{bmatrix} 0 & 0 & 5 \\ 7 & 2 & -5 \\ 3 & 1 & 8 \end{bmatrix} = 2(5)(7-6) = 10.$$

8. *Proof.* We prove the result by induction on the size of the matrix.

Base Case: Let $A = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$, then $\det(A) = ac$. Thus the result holds for 2×2 matrices.

Inductive Step: We assume that the determinant of an $n \times n$ upper triangular matrix is given by the product of the diagonal elements.

Let A be an $(n+1) \times (n+1)$ upper triangular matrix. We compute the determinant by expanding along the first column:

$$\det(A) = a_{11} \det(A_{11}) - a_{21} \det(A_{21}) + \cdots + (-1)^{n+1+1} a_{(n+1)1} \det(A_{(n+1)1})$$

Since A is upper-triangular, $a_{21} = \cdots = a_{(n+1)1} = 0$. Thus

$$\det(A) = a_{11} \det(A_{11}).$$

Since $A_1 1$ is an $n \times n$ upper triangular matrix, we can apply the induction hypothesis to get

$$\det(A) = a_{11} \det(A_{11}) = a_{11}(a_{22} \ldots a_{(n+1)(n+1)}).$$

Thus the result holds by mathematical induction. $\square$